





Retrieval Augmented Generation



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RAG quèsaco?

Overview

Introduction:

2022: ChatGPT is released and LLMs become a buzz word

What is a LLM ?

No official definition but they are distinguishable by their outstanding number of trainable parameters, often several billions, and the vast amount of data used for training them.

Maybe a relevant characteristics is that they are challenging to train and require a lot of resources.



Only a handful of players...

RAG quèsaco?

Overview



An interesting turn of events:

LlaMA is leaked on 4chan and a plethora of open-source LLMs are born starting with Alpaca which is trained on a dataset costing less than 500\$!

UNTIL March 2023

Building its own LLM:

When tailoring a LLM for domain-specific tasks two options prevail: Fine-Tuning (FT) a LLM or implementing a Retrieval-Augmented Generation (RAG) system.

RAG quèsaco?

Fine-tuning

Definition:

Fine-tuning involves training a pre-existing model on a specific dataset related to the target domain. This process allows the model to adapt and specialize in generating content relevant to the specified context.

FT a LLM:

Classical fine-tuning involves full fine-tuning. However, the resources required to fine-tune billions of parameters are prohibitive in addition to the inherent difficulty and instability of the training process.

Beyond full fine-tuning, several advanced variations exist, such as approaches focusing on updating only selected parameters. These methods are also known as Parameter Efficient Fine-Tuning techniques.

RAG quèsaco?

Fine-tuning

Limitations:

It remains a relatively resource-intensive, time-consuming, and expertise-demanding process, even when employing PEFT approaches.

Additionally, fine-tuning still grapples with limitations such as knowledge cutoff, i.e., considering new data necessitates a new training cycle to integrate this knowledge.

RAG quèsaco?

RAG

Retrieval Augmented Generation

RAG takes an input (a **query**) and retrieves a set of relevant/supporting documents given a source (e.g., Wikipedia). The documents are concatenated as **context** with the original input prompt and fed to the text generator which produces the final output.

It was introduced by Lewis et al. (Meta AI) in 2020:

(<https://arxiv.org/abs/2005.11401>)

arXiv:2005.11401v4 [cs.CL] 12 Apr 2021

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

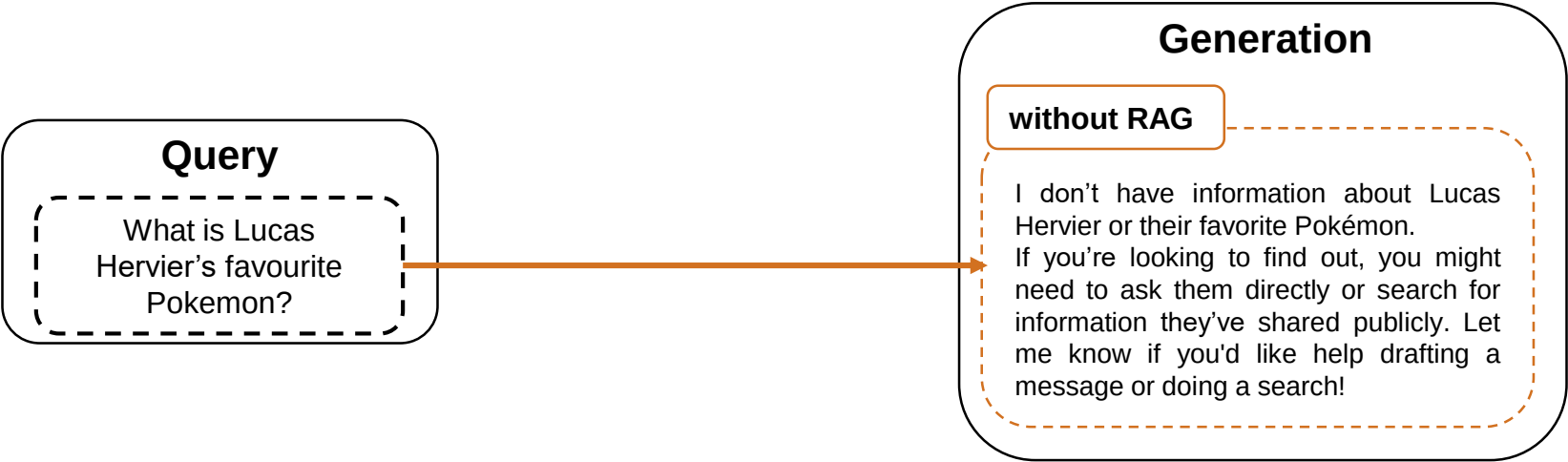
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Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. Pre-trained models with a differentiable access mechanism to explicit non-parametric memory have so far been only investigated for extractive downstream tasks. We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) — models which combine pre-trained parametric and non-parametric memory for language generation. We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We compare two RAG formulations, one which conditions on the same retrieved passages across the whole generated sequence, and another which can use different passages per token. We fine-tune and evaluate our models on a wide range of knowledge-intensive NLP tasks and set the state of the art on three open domain QA tasks, outperforming parametric seq2seq models and task-specific retrieve-and-extract architectures. For language generation tasks, we find that RAG models generate more specific, diverse and factual language than a state-of-the-art parametric-only seq2seq baseline.

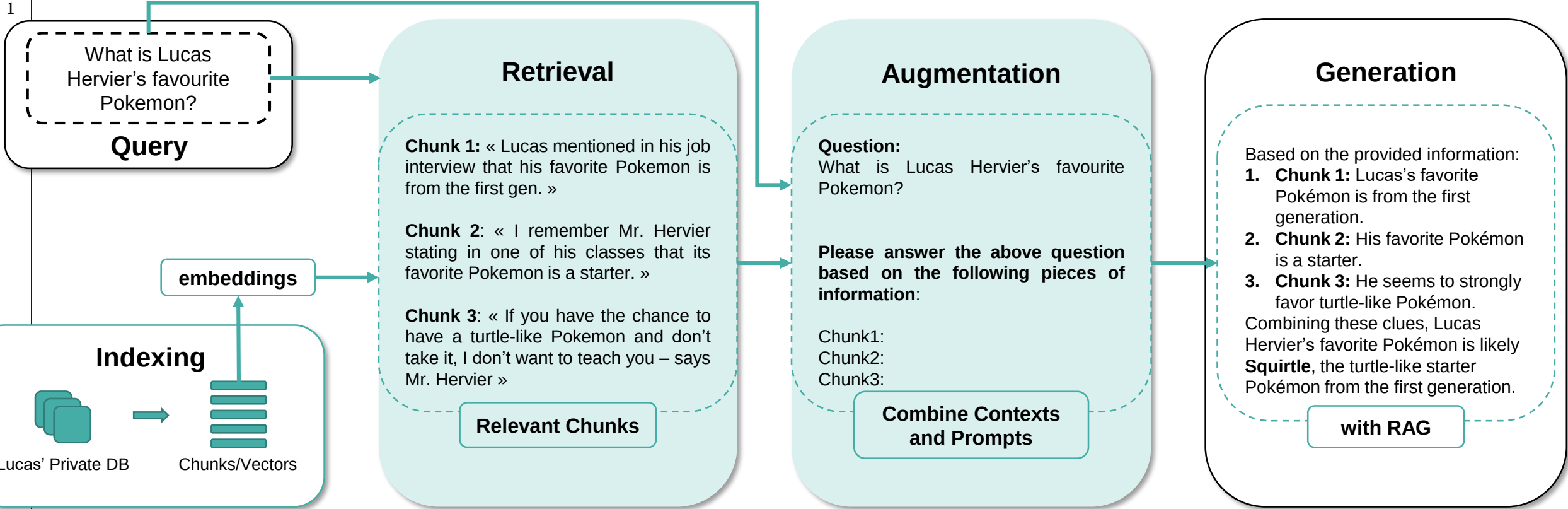
RAG quèsaco?

No RAG



RAG quèsaco?

RAG



RAG quèsaco?

RAG



Retrieval

Find documents or pieces of documents that are most likely connected and relevant to the query in a knowledge database.



Augmentation

Integrates the query with the retrieved data employing prompt engineering techniques. This step should effectively provide context to the LLM.



Generation

The generation step guarantees the model's output is coherent, accurate, and tailored according to the user's prompt.

RAG quèsaco?

RAG

Promises:

- Allows using up-to-date and/or proprietary data with LLMs
- Does not require re-training the LLMs
- Very plug-and-play system
- More specific/custom outputs
- Less « Hallucinations »
- Enhance « Trustworthiness »

RAG quèsaco?

RAG

A Magical Solution?

Not really. Many challenges arise when developing in-house RAG. The key takeaway being that building application-specific RAGs requires to specialize the different blocks of the system involving:

- Training or fine-tuning other models.
- Prompt Engineering.
- Custom retrievers.
- Dedicated parsers.
- ...

RAG quèsaco?

RAG vs FT



So Fine-Tuning or RAG ?

Source: "Retrieval-augmented generation for large language models: A survey", Gao et al., 2023

Feature Comparison	RAG	Fine-Tuning
Knowledge Updates	Directly updating the retrieval knowledge base ensures that the information remains current without the need for frequent retraining, making it well-suited for dynamic data environments.	Stores static data, requiring retraining for knowledge and data updates.
External Knowledge	Proficient in leveraging external resources, particularly suitable for accessing documents or other structured/unstructured databases.	Can be utilized to align the externally acquired knowledge from pretraining with large language models, but may be less practical for frequently changing data sources.
Data Processing	Involves minimal data processing and handling.	Depends on the creation of high-quality datasets, and limited datasets may not result in significant performance improvements.
Model Customization	Focuses on information retrieval and integrating external knowledge but may not fully customize model behavior or writing style.	Allows adjustments of LLM behavior, writing style, or specific domain knowledge based on specific tones or terms.
Interpretability	Responses can be traced back to specific data sources, providing higher interpretability and traceability.	Similar to a black box, it is not always clear why the model reacts a certain way, resulting in relatively lower interpretability.
Computational Resources	Depends on computational resources to support retrieval strategies and technologies related to databases. Additionally, it requires the maintenance of external data source integration and updates.	The preparation and curation of high-quality training datasets, defining fine-tuning objectives, and providing corresponding computational resources are necessary.
Latency Requirements	Involves data retrieval, which may lead to higher latency.	LLM after fine-tuning can respond without retrieval, resulting in lower latency.
Reducing Hallucinations	Inherently less prone to hallucinations as each answer is grounded in retrieved evidence.	Can help reduce hallucinations by training the model based on specific domain data but may still exhibit hallucinations when faced with unfamiliar input.
Ethical and Privacy Issues	Ethical and privacy concerns arise from the storage and retrieval of text from external databases.	Ethical and privacy concerns may arise due to sensitive content in the training data.



Evaluating a RAG system



Evaluating a RAG system

RAG a multi-components system

Numerous Components

As we have previously seen, a RAG system is a stack of different technical solutions: parsers, vectorizers, retrievers, prompt templates...

Which All Influence the System

Thus, the slightest modification to one of them impact the entire system!

For Example

We want to reduce the number of query' tokens so we only keep the 2 most-relevant chunks:

Evaluating a RAG system

RAG a multi-components system

Chunk 1: « Lucas mentioned in his job interview that his favorite Pokemon is from the first gen. »

Chunk 2: « I remember Mr. Hervier stating in one of his classes that its favorite Pokemon is a starter. »

Chunk 3: « If you have the chance to have a turtle-like Pokemon and don't take it, I don't want to teach you – says Mr. Hervier »

Question:
What is Lucas Hervier's favourite Pokemon?

Please answer the above question based on the following pieces of information:

Chunk 1
Chunk 2
Chunk 3



Question:
What is Lucas Hervier's favourite Pokemon?

Please answer the above question based on the following pieces of information:

Chunk 1
Chunk 2



Based on the provided information:

- Chunk 1:** Lucas's favorite Pokémon is from the first generation.
 - Chunk 2:** His favorite Pokémon is a starter.
 - Chunk 3:** He seems to strongly favor turtle-like Pokémon.
- Combining these clues, Lucas Hervier's favorite Pokémon is likely **Squirtle**, the turtle-like starter Pokémon from the first generation.



Based on the provided information:

- Chunk 1:** states that Lucas Hervier's favorite Pokémon is from the first generation.
 - Chunk 2:** states that his favorite Pokémon is a starter Pokémon.
- Combining these details, Lucas Hervier's favorite Pokémon is likely one of the first-generation starter Pokémon: **Bulbasaur, Charmander, or Squirtle**. However, the specific favorite is not explicitly mentioned in the chunks.

Evaluating a RAG system

RAG a multi-components system

Models keep on improving

Indeed, model providers are constantly updating new models. Thus, one should be able to easily and rapidly switch models to check if its improving the system for **its use-case!**

Other components evolve quickly too

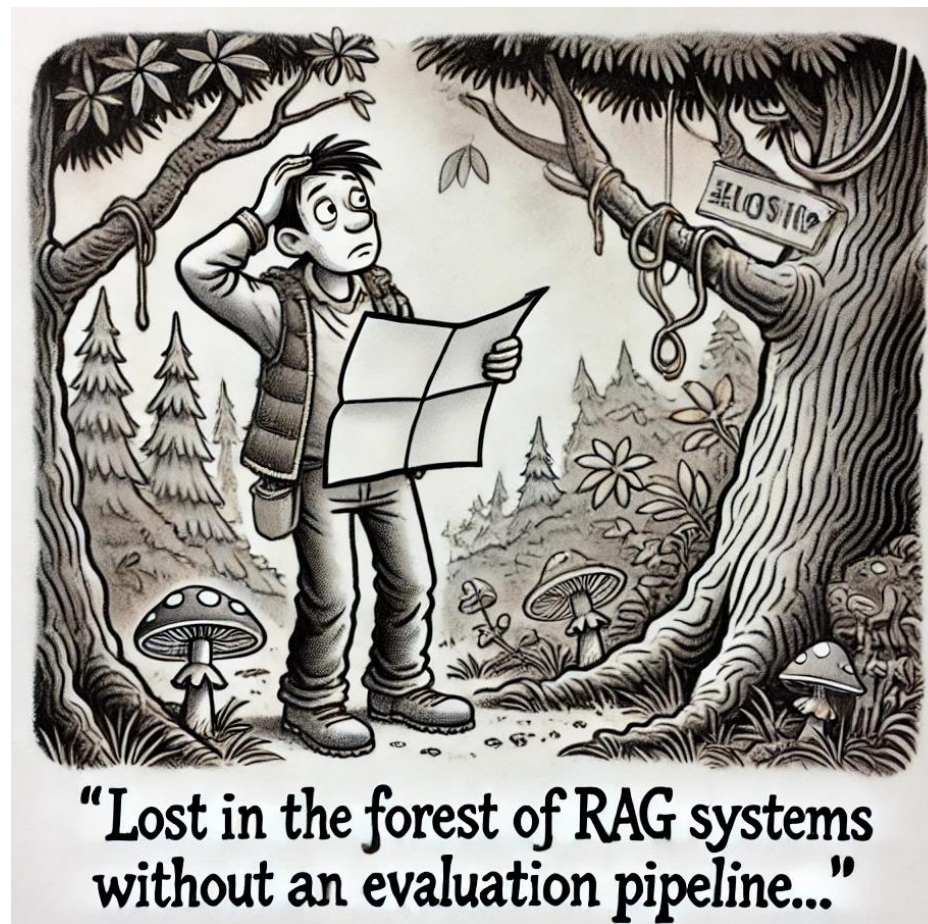
New retrievers strategy appears quickly, best practices are getting better and better, new prompt strategies emerge, etc... Consequently, each components is likely to be challenged!

The latest is not always the best

Just because a new components make your application looks good on a few examples does not mean it is better in general.

Evaluating a RAG system

RAG a multi-components system



Credit: ChatGPT 4o – 9th Jan. 2025

Evaluating a RAG system

Defining the evaluation

The Core Traits of an Effective Evaluation Framework



Representative

Correlates with expected outcomes, such as real-world performance.



Concise

Summarized with only a few key metrics, avoiding unnecessary complexity.



Speed

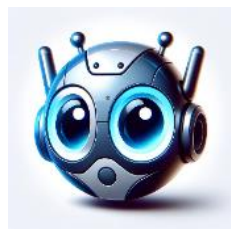
Enables fast and automated iterations to keep the process efficient..

Source: [Josh Tobin's talk on YouTube - Evaluating LLM-based Applications](#)

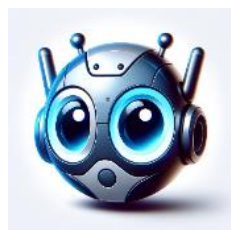
Evaluating a RAG system

Defining the evaluation

Classification Task



DOG



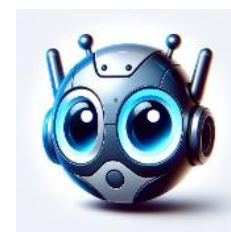
CAT

Accuracy: 50%

Summarization Task



Can you please summarize Harry Potter in only 1 short sentence?



A young wizard, Harry Potter, confronts a dark sorcerer while discovering the power of friendship, love, and bravery.

Score: ???

Evaluating a RAG system

Defining the evaluation



**We struggle to evaluate GenAI applications as they often requires
SUBJECTIVE EVALUATION *i.e.* QUALITATIVE METRICS**

Evaluating a RAG system

Defining the evaluation

LLMs (and SLMs) rarely focus on a single task



Can you summarize ...

Summarization



$1 + 1 = ?$

Calculus



Is the author happy?

Sentiment Analysis



Propose code to...

Coding



HARD to summarize with a few metrics

Evaluating a RAG system

Defining the evaluation

Evaluation Set

Q: Who is Ron Weasley?

Q: What is a Dementor?

Q: How can you win a Quidditch match?

Harry Potter

Production Set

Q: Who is Samwise Gamgee?

Q: What is an Istari?

Q: How can you reclaim the One Ring?

THE LORD OF THE RINGS

There is ALWAYS DATA DRIFT!

Evaluating a RAG system

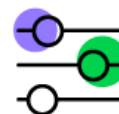
Defining the evaluation

Key Challenges in Evaluating GenAI Applications



Subjectivity

Often relies on qualitative metrics rather than quantitative ones, making evaluations less objective.



Task Diversity

GenAI models handle diverse, complex tasks, which complicates standardized evaluation.



Drift

Users challenge the model in unexpected domains, revealing gaps in its training.

Inspiration: [Josh Tobin's talk on YouTube - Evaluating LLM-based Applications](#)

Evaluating a RAG system

Defining the evaluation

The Core Traits of an Effective Evaluation Framework



Representative



Precision



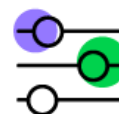
Speed



Key Challenges in Evaluating Generative Applications



Subjectivity



Task Diversity



Drift

Evaluating a RAG system

How Helpful Are Public Benchmarks?

Functional Correctness

Examples: HumanEval, StaticEval

What It Measures: Whether the model output solves a specific task (e.g., code completion).

Key Insight: Actually useful if you care about the downstream task.

Live Human Eval

Examples: Chatbot arena, HALIE

What It Measures: Head-to-head performance where users prompt models and select the better output.

Key Insight: Tempting to see it as the gold standard, but prone to biases and subjective flaws.

Source: [Josh Tobin's talk on YouTube - Evaluating LLM-based Applications](#)

Models Evaluating Models

Examples: AlpacaEval, GEMBA

What It Measures: Benchmarks where one model evaluates another's output.

Key Insight: Increasingly popular but requires caution in interpreting results.

Holistic Benchmarks

Examples: HeLM, CoTHub, BigBench

What It Measures: Broad benchmarks that cover a wide range of tasks.

Key Insight: Appears comprehensive but might miss your specific task.

Matching Human Answers

Examples: BLEU, ROUGE

What It Measures: Heuristic-based metrics for comparing model outputs to human references.

Key Insight: Quick and easy to compute but outdated, with biases making them less reliable today.

Evaluating a RAG system

How Helpful Are Public Benchmarks?



THEY DO NOT COVER ONE'S USE-CASE!

- Public benchmarks are useful for gaining a rough comparison of different solutions.
- However, they must be validated and tailored to one's specific pipeline for reliable insights.

Evaluating a RAG system

Building an evaluation pipeline

Ensure the highest possible COVERAGE!



Formalize

Spend time to formalize your use case's outputs and the task(s) the model is supposed to solve!



Iterate

It will (should) be an iterative process. One should involve as many contributors as possible.



Collect

One needs to think on how it will collect the user's (production) data and its feedbacks.

Evaluating a RAG system

Building an evaluation pipeline - Metrics

Quantitative Metrics

These metrics provide **measurable, objective insights** that can be directly compared across models or iterations. They focus on **numerical values**, making them **reliable** for tracking progress and optimizing performance over time. Additionally, they tend to **scale well** and are **robust** due to their **repeatability**.

However, they are **not universally applicable** and **may lack exhaustiveness**, especially in capturing subjective or nuanced aspects of model performance.



Evaluating a RAG system

Building an evaluation pipeline - Metrics

Pseudo – Quantitative Metrics

These metrics bridge the gap between quantitative precision and semantic understanding by **evaluating content overlap, contextual meaning, or both**. They are **scalable, flexible, and effective** for tasks like text generation and summarization **where exact matches are not feasible**.

However, their performance **heavily depends on the task context, reference quality, and underlying models**, which can introduce biases. While **practical and automated**, these metrics may **not fully capture fluency or align with human judgment**, making them best used alongside other evaluation methods.

ROUGE

Recall-Oriented Understudy for Gisting Evaluation (ROUGE) is a set of metrics designed to evaluate the quality of summaries by comparing them to human-generated reference summaries.

BLEU

Bilingual Evaluation Understudy (BLEU) is a metric for evaluating the quality of machine-translated text against one or more reference translations.

Semantic Similarity

The core idea is to use an embedding model (e.g., Sentence-BERT) to generate embedding vectors for texts and then calculate their similarity using a metric such as cosine similarity.

Evaluating a RAG system

Building an evaluation pipeline - Metrics



Human Evaluation

Human evaluation uses human judges to assess a model's performance based on specific criteria, providing **qualitative feedback** based on their understanding and expectations of the task. It is considered **the gold standard** for evaluating language models.

However, human evaluation is not a silver bullet, as annotators' **subjectivity significantly influences the results**. Additionally, such evaluations **scale poorly** and can become **prohibitively expensive** over time.

Likert Scales

Squirtle is, without any doubt, the best starter in all Pokemon's games:

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

Pairwise Comparison

Choose the most accurate response:

Model A

Dogs are better than cats.

Model B

Cats are better than dogs.

Evaluating a RAG system

Building an evaluation pipeline - Metrics

LLM Evaluation

LLM evaluation involves using LLM to assess the outputs of other models based on specific criteria. LLMs provide **AUTOMATED qualitative feedback**. This approach is **scalable and cost-efficient compared to human evaluation**.

However, it is not without limitations: LLMs can **exhibit biases**, **may not align with human judgment**, and could **overestimate performance** when evaluating similar models.

Sources:

[Large Language Models are not Fair Evaluators](#)

[G-EVAL](#)

[Josh Tobin's talk on YouTube - Evaluating LLM-based Applications](#)



Evaluation

System Prompt

You are an expert evaluator tasked with assessing the quality of model-generated outputs. Evaluate the given response based on the following criterion: relevance to the input prompt. Assign a score from 1 to 5, and provide a brief explanation for your rating. Be objective and precise in your evaluation.

Input Prompt

Who is the best Pokémon starter in the first generation?

Model-Generated Response

The best starter Pokémon is definitely Pikachu as it is Ash's companion and Ash finally became the very best.

Evaluation

- **Relevance to the Input Prompt: 2/5**

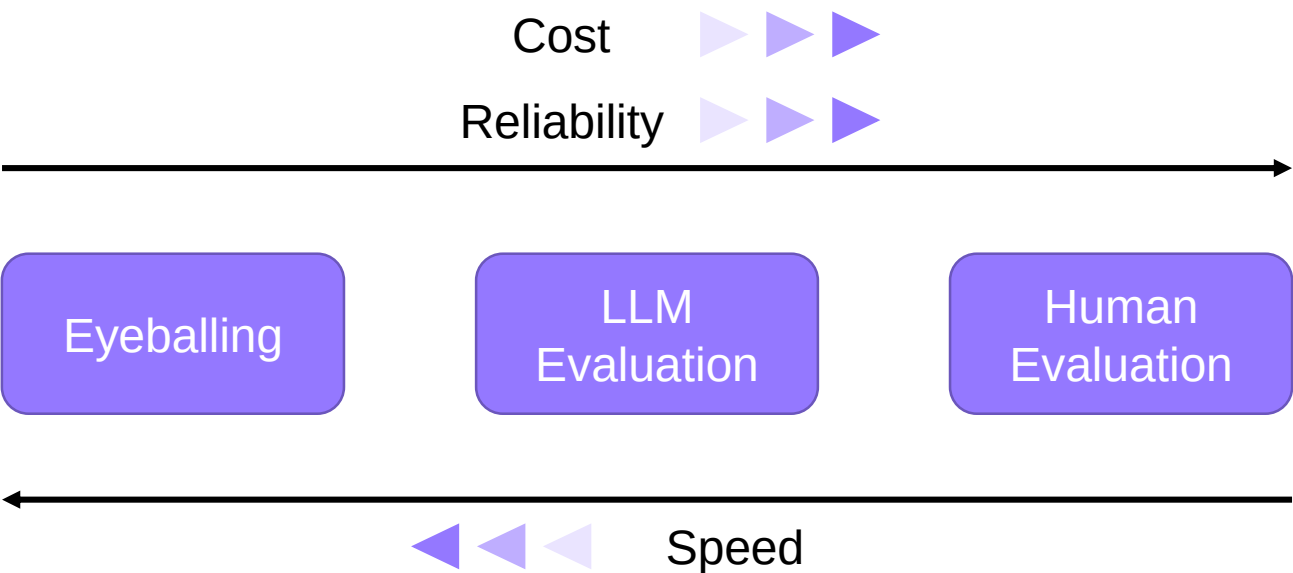
Explanation: The response mentions a well-known Pokémon (Pikachu) but fails to address the specific context of the question. Pikachu is not one of the designated starter Pokémon in the first generation (Bulbasaur, Charmander, and Squirtle are). The mention of Ash and his journey, while related to the Pokémon franchise, is irrelevant to the prompt's focus on starter Pokémon in the games.

Evaluating a RAG system

Building an evaluation pipeline - Metrics



Qualitative Assessment Tradeoffs



Source: [W&B course – RAG++: From POC to Production](#)

Evaluating a RAG system

Building an evaluation pipeline - Metrics

Balancing Cost, Speed, and Efficiency in Human-Verified LLM Evaluation

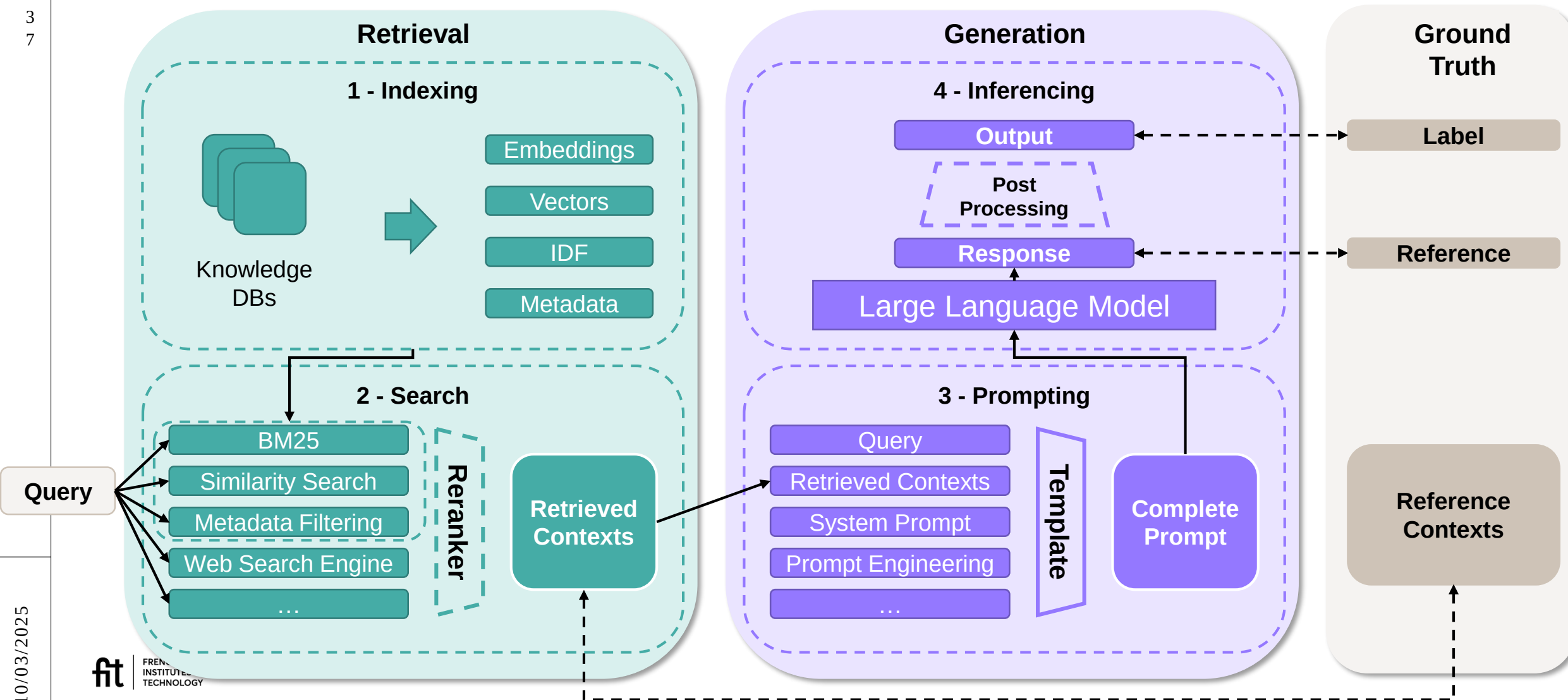


Source: [Josh Tobin's talk on YouTube - Evaluating LLM-based Applications](#)

Credit Images: ChatGPT 4o

Evaluating a RAG system

Building an evaluation pipeline – Component Wise

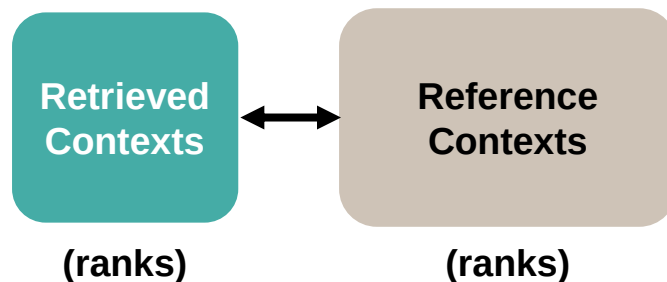


Evaluating a RAG system

Building an evaluation pipeline – Component Wise

Evaluating Retrieval

Accuracy



Measure the system's ability to find relevant context elements in comparison to a set of reference contexts. It can also incorporate the notion of rank.

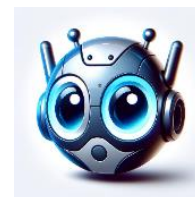
Evaluation Approaches:



Quantitative Metrics



Pseudo-Quantitative



LLM Evaluation



Human Evaluation

Evaluating a RAG system

Building an evaluation pipeline – Component Wise

Evaluating Retrieval

Accuracy – (Some) Quantitative Metrics

TP: True positives
TN: True Negatives
RC: Relevant Contexts
Top_k: Top-k Retrieved Contexts
Q: Total number of queries

rel(k): Indicator function equaling 1 if the item at rank *k* is a relevant document
rank_q: Rank position of the first relevant document for the *q*-th query

Non-Rank Based

Accuracy

$$\frac{TP + TN}{|RC|}$$

Precision@k

$$\frac{|RC \cap Top_k|}{k}$$

Recall@k

$$\frac{|RC \cap Top_k|}{|RC|}$$

Rank Based

AP@K

$$\frac{1}{|RC|} \sum_{k=1}^K Precision@k \times rel(k)$$

MAP@K

$$\frac{1}{Q} \sum_{q=1}^Q AP@K_q$$

MRR

$$\frac{1}{Q} \sum_{q=1}^Q \frac{1}{rank_q}$$

Evaluating a RAG system

Building an evaluation pipeline – Component Wise

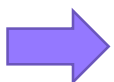
Evaluating Retrieval

Accuracy – (Some) Quantitative Metrics



Sometimes, **Exact Matches Are Not Feasible**:

- Segmentation Differences
- Paraphrasing, Rephrasing
- Duplicates of Reference Contexts in Different Documents
- Lack of exhaustivity
- ...



Relaxed Matching, Pseudo-Quantitative metrics (e.g. **Semantic Similarity**)

Evaluating a RAG system

Building an evaluation pipeline – Component Wise

Evaluating Retrieval

Accuracy – LLM Evaluation

```
RETRIEVAL_ACCURACY_PROMPT = """
For each retrieved context, compare it to the reference context and classify the information as either:
- Matched: The retrieved context accurately reflects part of the reference.
- Unmatched: The retrieved context does not align with any part of the reference.
Based on the matched information, calculate the proportion of the reference context that is covered by
the retrieved contexts and express this as a percentage. Briefly explain any missing or irrelevant parts.

### Retrieved Contexts
{{retrieved_contexts}}

### Reference Contexts
{{reference_contexts}}

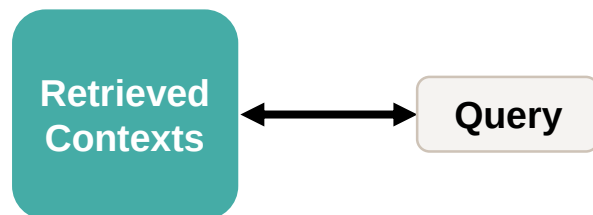
### Response
"""
```


Evaluating a RAG system

Building an evaluation pipeline – Component Wise

Evaluating Retrieval

Relevance

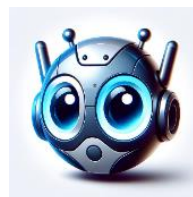


Evaluates how well the retrieved documents match the information needed expressed in the query. It measures the precision and specificity of the retrieval process.

Evaluation Approaches:



Pseudo-Quantitative



LLM Evaluation



Human Evaluation

Evaluating a RAG system

Building an evaluation pipeline – Component Wise

Evaluating Retrieval

Relevance – LLM Evaluation



```
RETRIEVAL_RELEVANCE_PROMPT = ""
```

```
You are an expert evaluator tasked with assessing the relevance of retrieved documents to a given query.
Your role is to determine how well the content of each retrieved document matches the information needs
expressed in the query. Specifically:
```

```
Precision: Does the document contain information directly related to the query?
```

```
Specificity: Is the information detailed and focused on the query topic?
```

```
For each retrieved document, provide the following:
```

```
- Relevance Classification: Mark the document as "Highly Relevant," "Somewhat Relevant," or "Irrelevant."
```

```
- Explanation: Briefly explain your classification, highlighting any relevant or irrelevant details.
```

```
### Query
```

```
{{query}}
```

```
### Retrieved Contexts
```

```
{{retrieved_contexts}}
```

```
### Response
```

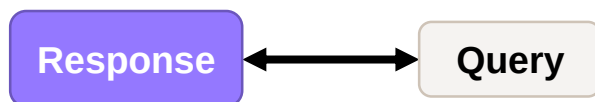
```
""
```

Evaluating a RAG system

Building an evaluation pipeline – Component Wise

Evaluating Generation

Relevance

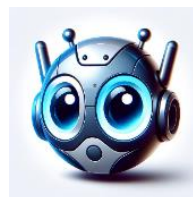


Measures how well the generated response aligns with the intent and content of the initial query. It ensures that the response is related to the query topic and meets the query's specific requirements.

Evaluation Approaches:



Pseudo-Quantitative



LLM Evaluation



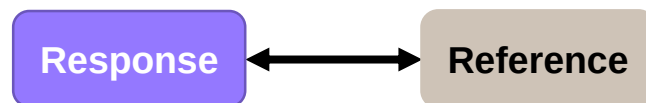
Human Evaluation

Evaluating a RAG system

Building an evaluation pipeline – Component Wise

Evaluating Generation

Correctness

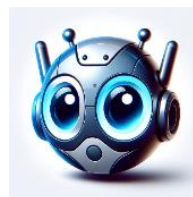


Measures the accuracy of the generated response against a sample response, which serves as a ground truth. It checks if the response is correct in terms of factual information

Evaluation Approaches:



Pseudo-Quantitative



LLM Evaluation



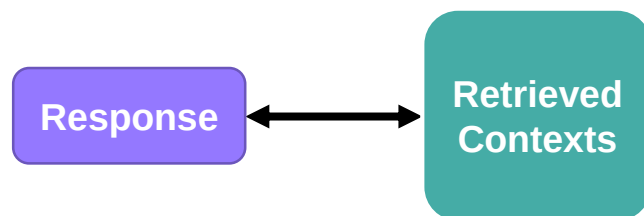
Human Evaluation

Evaluating a RAG system

Building an evaluation pipeline – Component Wise

Evaluating Generation

Faithfulness

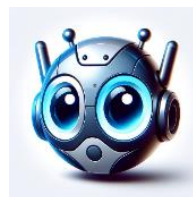


Evaluate whether the generated response accurately reflects the information contained within the retrieved documents and measure the consistency between the generated content and the source documents.

Evaluation Approaches:



Pseudo-Quantitative



LLM Evaluation



Human Evaluation

Evaluating a RAG system

Building an evaluation pipeline – End-to-End

Evaluating the Whole System

What is End-to-End Evaluation?

End-to-end evaluation measures the overall performance of a system by focusing on its ability to achieve the desired outcomes, typically assessed through the relationship between the input (e.g., query) and the output (e.g., response or label).

Unlike component-level evaluation, it does not analyze or account for the interactions or dependencies within the system's intermediate steps.

Why is it important?

RAG systems are inherently complex, and “the interplay between the retrieval and generation components means that the entire system’s performance cannot be fully understood by evaluating each component in isolation [1,2]”.

Source: Hao Yu et al., [Evaluation of Retrieval-Augmented Generation: A Survey](#), 2024

[1]: Saad-Falcon et al., [ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems](#), 2023

[2]: James et al., [Ragas: Automated evaluation of retrieval augmented generation](#), 2023

Evaluating a RAG system

Building an evaluation pipeline



Conclusion

Ultimately, **what should guide you is your end-to-end evaluation**, as improving one component in isolation can sometimes lead to a deterioration of the overall system performance.

Beyond performance metrics, it's important to also consider other critical factors such as:

- **Latency**: How quickly the system responds.
- **Cost**: The computational or financial expense of running the system.
- **Toxicity**: The system's ability to avoid generating harmful or offensive content.
- **Other Domain-Specific Metrics**: Depending on your use case, additional metrics may be relevant.
- ...



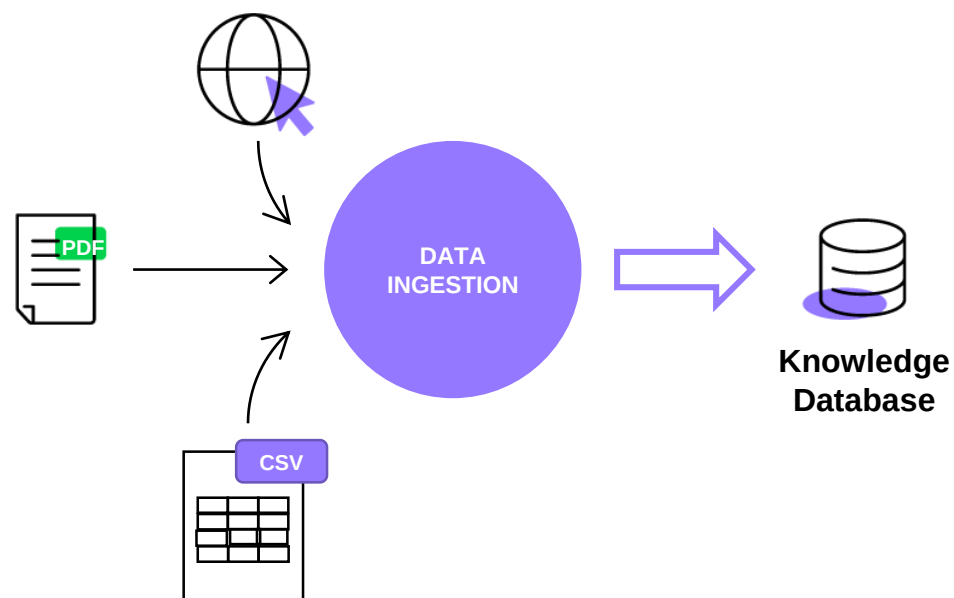
Data Ingestion



Data Ingestion

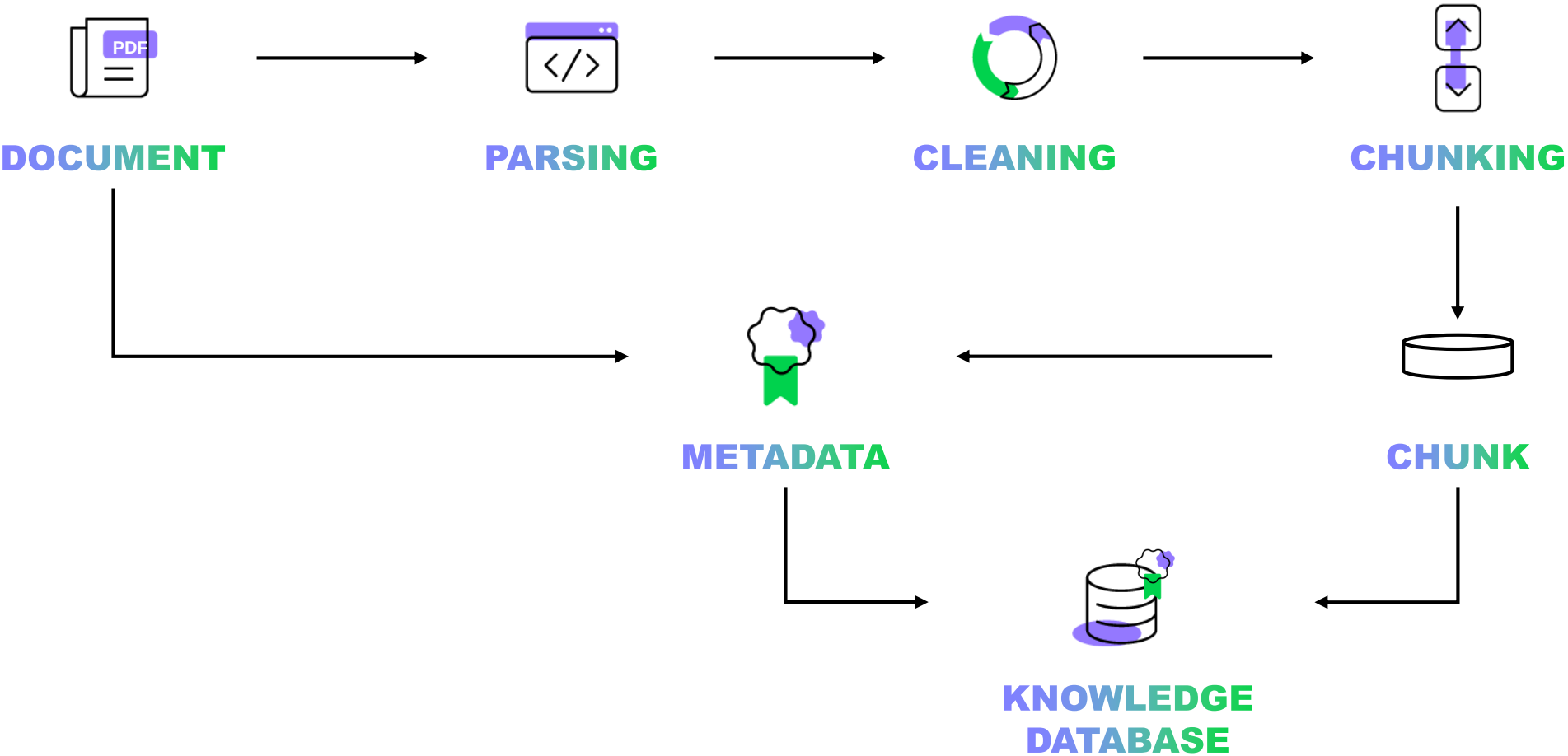
Introduction

Data ingestion is the comprehensive process of **collecting, processing, and preparing raw documents for analysis and retrieval**. In RAG applications, effective ingestion ensures that the **system operates on high-quality, structured data**.



Data Ingestion

Introduction

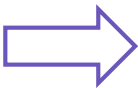
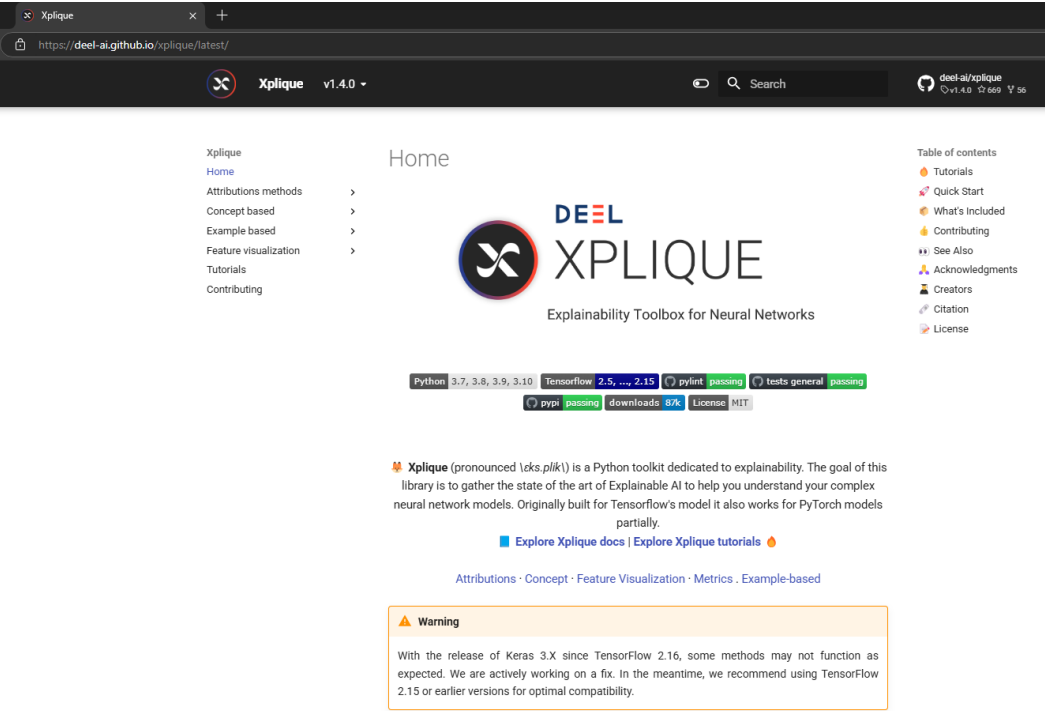


Data Ingestion

Parsing

Definition

Data parsing is the first step where raw documents —such as PDFs, web pages, or scanned images— are extracted and transformed into machine-readable format.



```
<!doctype html>
<html lang="en" class="no-js">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width,initial-scale=1">
    <link rel="next" href="api/attributions/api_attributions/">
    <link rel="icon" href="assets/x_logo.png">
    <meta name="generator" content="mkdocs-1.6.1, mkdocs-material-9.5.39">
    <title>Xplique</title>
    <link rel="stylesheet" href="assets/stylesheets/main.8c3ca2c6.min.css">
    <link rel="stylesheet" href="assets/stylesheets/palette.06af60db.min.css">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link rel="stylesheet" href="https://fonts.googleapis.com/css?family=Roboto:300,300i,400,400i,700,700i%7CRoboto+Mono:400,400i,700,700i&display=fallback">
    <style>:root{--md-text-font:"Roboto";--md-code-font:"Roboto Mono"}</style>
    <link rel="stylesheet" href="css/custom.css">
    <script>__md_scope=new URL(".",location),__md_hash=e=>[...e].reduce(((e,_)=>(e<<5)-e+_.charCodeAt(0)),0),__md_get=(e,_,localStorage,t=__md_scope)>JSON.parse(_.getItem(t.pathname+"."+e)),__md_set=(e,_,t=localStorage,a=__md_scope)>{try{t.setItem(a.pathname+"."+e,JSON.stringify(_))}catch(e){}}</script>
  </head>
  <body dir="ltr" data-md-color-scheme="default" data-md-color-primary="dark" data-md-color-accent="indigo">
    <input class="md-toggle" data-md-toggle="drawer" type="checkbox" id="__drawer" autocomplete="off">
    <input class="md-toggle" data-md-toggle="search" type="checkbox" id="__search" autocomplete="off">
    <label class="md-overlay" for="__drawer"></label>
    <div data-md-component="skip">
      <a href="#tutorials" class="md-skip">
        Skip to content
      </a>
    </div>
    <div data-md-component="announce">
    </div>
    <div data-md-color-scheme="default" data-md-component="outdated" hidden>
      <aside class="md-banner md-banner--warning">
        <div class="md-banner__inner md-grid md-typeset">
          You're not viewing the latest version.
          <a href="..">
            <strong>Click here to go to latest.</strong>
          </a>
        </div>
      </aside>
    </div>
  </body>
</html>
```

Data Ingestion

Cleaning

Definition

Data cleaning refines the parsed document by removing noise, such as irrelevant formatting, headers, footers, and typographical errors. It can also include pre-processing steps such as consistent casing, punctuation standardization, etc...

```
<!doctype html>
<html lang="en" class="no-js">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width,initial-scale=1">
    <link rel="next" href="api/attributions/api_attributions/">
    <link rel="icon" href="assets/x_logo.png">
    <meta name="generator" content="mkdocs-1.6.1, mkdocs-material-9.5.39">
    <title>Xplique</title>
    <link rel="stylesheet" href="assets/stylesheets/main.8c3ca2c6.min.css">
    <link rel="stylesheet" href="assets/stylesheets/palette.06af60db.min.css">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link rel="stylesheet" href="https://fonts.googleapis.com/css?
family=Roboto:300,300i,400,400i,700,700i%7CRoboto+Mono:400,400i,700,700i&display=fallback">
    <style>:root{--md-text-font:"Roboto";--md-code-font:"Roboto Mono"}</style>
    <link rel="stylesheet" href="css/custom.css">
    <script>_md_scope=new URL(".",location),_md_hash=e=>[...e].reduce(((e,_)=>{e<<5)-
e+_.charCodeAt(0)),0),_md_get=
(e,_=localStorage,t=_md_scope)=JSON.parse(_.getItem(t.pathname+"."+e)),_md_set=
(e,_,t=localStorage,a=_md_scope)=>{try{t.setItem(a.pathname+"."+e,JSON.stringify(_))}catch(e){}}
</script>
  </head>
  <body dir="ltr" data-md-color-scheme="default" data-md-color-primary="dark" data-md-color-
accent="indigo">
    <input class="md-toggle" data-md-toggle="drawer" type="checkbox" id="__drawer" autocomplete="off">
    <input class="md-toggle" data-md-toggle="search" type="checkbox" id="__search" autocomplete="off">
    <label class="md-overlay" for="__drawer"></label>
    <div data-md-component="skip">
      <a href="#tutorials" class="md-skip">
        Skip to content
      </a>
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    </div>
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      <aside class="md-banner md-banner--warning">
        <div class="md-banner__inner md-grid md-typeset">
          You're not viewing the latest version.
          <a href="..">
            <strong>Click here to go to latest.</strong>
          </a>
        </div>
      </aside>
    </div>
```



```
# Home
Xplique
```

Xplique (pronounced /ɛks.plik/) is a Python toolkit dedicated to explainability. The goal of this library is to gather the state of the art of Explainable AI to help you understand your complex neural network models. Originally built for TensorFlow's model it also works for PyTorch models partially.

Explore Xplique docs | Explore Xplique tutorials
Attributions · Concept · Feature Visualization · Metrics · Example-based

!! Warning
With the release of Keras 3.X since TensorFlow 2.16, some methods may not function as expected. We are actively working on a fix. In the meantime, we recommend using TensorFlow 2.15 or earlier versions for optimal compatibility.

The library is composed of several modules, the Attributions Methods module implements various methods (e.g Saliency, Grad-CAM, Integrated-Gradients...), with explanations, examples and links to official papers. The Feature Visualization module allows to see how neural networks build their understanding of images by finding inputs that maximize neurons, channels, layers or compositions of these elements. The Concepts module allows you to extract human concepts from a model and to test their usefulness with respect to a class. Finally, the Metrics module covers the current metrics used in explainability. Used in conjunction with the Attribution Methods module, it allows you to test the different methods or evaluate the explanations of a model.

Data Ingestion

Chunking

Definition

Chunking in a RAG pipeline is a critical process that involves **breaking down a large, unstructured document into smaller, coherent segments or "chunks"** that balance the preservation of context with efficient retrieval.

Why does one chunk?

1. Respect Context Window Limits
2. Improve Retrieval Precision and Efficiency
3. Avoid the “Needle in the Haystack” Problem

How to chunk?

A wide range of chunking strategies exists and their efficiency and relevance **depends on the use case and the nature of the documents**. Common strategies include: Fixed-Token Splitting, Sentence-level chunking, Recursive chunking or Semantic Chunking.

Data Ingestion

Chunking

Build with the Space: https://huggingface.co/spaces/m-ric/chunk_visualizer



Method to split chunks 📄

🔗 LangChain's CharacterTextSplitter

Separators used in RecursiveCharacterTextSplitter

```
['\\n#{1,6} ', '```\n', '\\n\\|*\\|*\\|*+\\n', '\\n---+\\n', '\\n__+\\n', '\\n\\n', '\\n', ' ', '']
```

Choose a preset
This will apply a specific set of separators to RecursiveCharacterTextSplitter.

Default

Python

Markdown

Length function
How should we measure our chunk lengths?

Token count (BERT tokens)

Chunk length 📏
In the chosen unit.
50 500

Overlap between chunks
In the chosen unit.
0 50

🔊 Output

Overlap

Chunk 0

Chunk 1

Chunk 2

Chunk 4

Chunk 6

Chunk 8

Chunk 9

Chunk 10

Chunk 12

Chunk 14

Chunk 16

Chunk 17

Chunk 18

Chunk 19

Chunk 21

Chunk 22

Chunk 24

Chunk 25

Chunk 26

Chunk 27

Chunk 28

Chunk 29

Chunk 30

Chunk 31

Chunk 32

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`['\n#{1,6} ', '```\n', '\n***+\\n', '\n---+\\n', '\n__+\\n', '\n\\n', '\n', ' ', '']`

Choose a preset
This will apply a specific set of separators to RecursiveCharacterTextSplitter.

Default

Python

Markdown

Length function
How should we measure our chunk lengths?

Token count (BERT tokens)

Chunk length 📏
In the chosen unit.
130 🔄
50 500

Overlap between chunks
In the chosen unit.
10 🔄
0 50

🔊 Output

Overlap

Chunk 0

Chunk 1

Chunk 2

Chunk 4

Chunk 6

Chunk 8

Chunk 9

Chunk 10

Chunk 12

Chunk 14

Chunk 16

Chunk 17

Chunk 18

Chunk 19

Chunk 21

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Chunk 29

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Chunk 31

Chunk 32

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Data Ingestion

Chunking

Semantic Chunking

It is a refined strategy that has proven effective in real-world scenarios. It first splits text into sentences then **groups those sentences according to semantic similarity**. It offers several advantages:

- **Improved Relevance**: Clustering thematically related sentences ensures that each chunk contains a complete set of relevant information.
- **Context Preservation**: Logical relationships remain intact, which is vital for accurate comprehension and generation by large language models.
- **Adaptive Segmentation**: Chunk sizes adapt to the text's natural structure, creating more coherent segments than a one-size-fits-all.

However, it often **represents a computational overhead**, so it is recommended not to use it as a baseline.

For further details on how this method was evaluated, consult this research article [2] on chunking.

Source:

[1]: <https://github.com/wandb/edu/blob/main/rag-advanced/notebooks/Chapter03.ipynb>

[2]: <https://research.trychroma.com/evaluating-chunking>

Data Ingestion

Metadata

Goal

Attaching metadata to each chunk adds valuable context during retrieval and generation. This information **makes it easier to organize, filter, and prioritize chunks**, and it helps trace each chunk back to its source.

Examples

- source document
- chunk position
- unique chunk ID
- creation time
- section or chapter reference
- summary of the section
- author
- topic or category

Source:

[1]: <https://www.multimodal.dev/post/how-to-chunk-documents-for-rag?ref=chitika.com>



Retrieval Strategies



Retrieval Strategies

Sparse (Lexical) Retrieval

Definition

Lexical retrieval methods match queries and documents based on exact word occurrences using sparse vector representations.

Key Techniques

1. **TF-IDF:** Assigns importance to words based on frequency across documents.
2. **BM25:** A probabilistic ranking function improving TF-IDF with term saturation and document length normalization.

Advantages

- Highly efficient and interpretable.
- Works well for keyword-based search.

Limitations

- No semantic understanding (e.g. “car” ≠ “vehicle”)
- Struggle with paraphrases

Retrieval Strategies



Sparse (Lexical) Retrieval

Corpus

- Document 1: “Cyber threats pose a risk to organizations.”
- Document 2: “Inadequate cyber defenses expose organizations to cyber risks.”

Vocabulary (after tokenization and sorting by alphabetical order)

[cyber, defense, expose, inadequate, organization, pose, risk, threats]

TF-IDF of Document 2 (with the Corpus & Vocabulary)

	cyber	defense	expose	inadequate	organization	pose	risk	threats
# docs w/ the token	2	1	1	1	2	1	2	1
IDF	$\log(\frac{2}{2})$	$\log(\frac{2}{1})$	$\log(\frac{2}{1})$	$\log(\frac{2}{1})$	$\log(\frac{2}{2})$	$\log(\frac{2}{1})$	$\log(\frac{2}{2})$	$\log(\frac{2}{1})$
TF(*,Doc2)	$\frac{2}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$
TF - IDF(*,Doc2)	0	0.043	0.043	0.043	0	0.043	0	0

Retrieval Strategies

Dense Retrieval

Definition

Dense retrieval **represents queries and documents as dense embeddings** (vector representations), allowing for **semantic similarity search** rather than exact keyword matching.

Key Techniques

1. **BERT-based models:** Learn contextual embeddings.
2. **Sentence Transformers:** Optimize for sentence-level similarity
3. **Multi-modal models** (e.g., CLIP for text-image retrieval)

Advantages

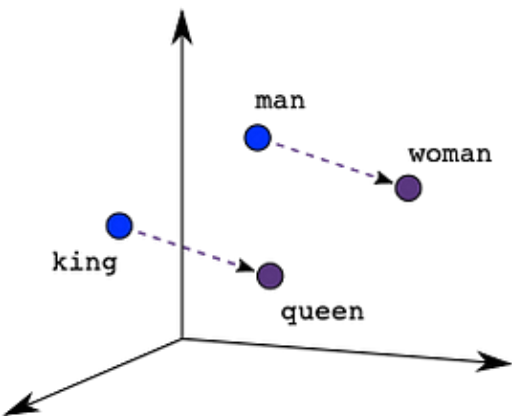
- Captures synonyms and meaning beyond exact words.
- Works well for open-ended and conversational queries.

Limitations

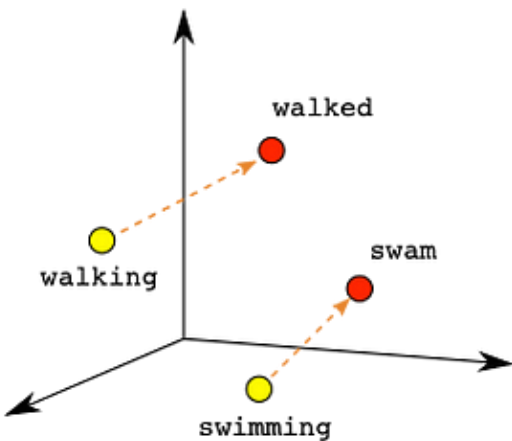
- Requires more computation.
- Less interpretable than BM25.

Retrieval Strategies

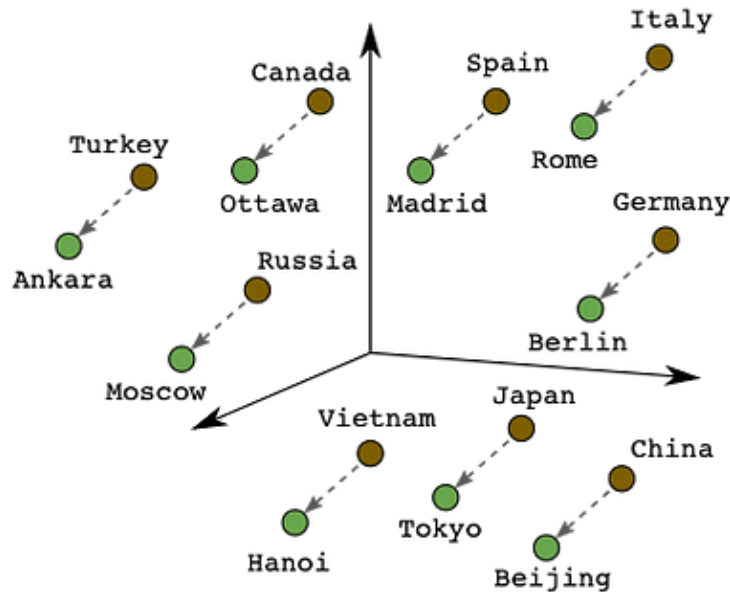
Dense Retrieval



Male-Female



Verb Tense



Country-Capital

Linear Relationships between Words in an Embedding space. Image from developer.google.com

Retrieval Strategies

Hybrid Retrieval

Definition

Hybrid (or ensemble) retrieval **combines multiple retrieval strategies** (often keywords and semantic search) to improve recall and precision. Technically, it consists of a fusion algorithm.

Key Techniques

1. **Score-Based Fusion:** Normalizes scores from multiple retrievers and combine them.
2. **Rank-Based Fusion:** Merges ranked lists from different retrievers using techniques like Reciprocal Rank Fusion (RRF).
3. **Late Fusion:** Retrieves results separately then apply a re-ranking algorithm.

Advantages

- Combines exact matching (lexical) and semantic understanding (dense)
- Works well for both keyword-based and open-ended queries.

Limitations

- Requires more computation and storage.
- Complex Fusion & Interpretability Issues.



Pre/Post-Retrieval



Pre-Retrieval

Query Optimization

Purpose

Formulating a precise and clear query is challenging, and poorly formulated queries lead to suboptimal retrieval effectiveness. Consequently, applying query optimization techniques enhances the performance of our RAG system.

Query expansion

- **Idea:** Expanding the query into multiple queries to enrich context.
- **Example:** Generating sub-queries that break down the original question into necessary sub-questions, ensuring a more comprehensive and contextualized retrieval.

Query Transformation

- **Idea:** Retrieving relevant text chunks based on a transformed query instead of the user's original query.
- **Examples:** Query rewriting using LLMs (see example next slide), fixing mistakes

Query Routing

- **Idea:** Dynamically routing different types of queries to the most suitable RAG pipeline to accommodate diverse retrieval needs.
- **Example:** Metadata Filtering, i.e. extracting key query terms, followed by filtering documents based on both keywords and metadata attributes.

Pre-Retrieval

Query Optimization

Imagine a new pokemon

Let's imagine a new Pokémon that's a fusion of two classic elements: fire and ice. This Pokémon, let's call it "Frostflame," embodies the fierce energy of a blaze and the serene chill of ice. It has a sleek, dragon-like body with scales that shimmer in a gradient from deep blue at its tail to bright orange at its head. Its wings are large and feathered, half covered in flames and half in frost, allowing it to control both elements in battle. Its eyes glow with a fiery orange light, while its breath can either scorch the air or freeze it solid. Frostflame's tail ends in a crystalline ice shard that's constantly surrounded by a small, swirling flame, symbolizing its dual nature.

Now, I'll create an image of Frostflame, the fire and ice Pokémon, using these detailed characteristics.

It looks like there was an issue with creating the image. Let me try that again for you.



Post-Retrieval

The Challenge: Overloaded Context & Lost Information

- Excessive retrieved information **can reduce the effectiveness** of LLM-generated responses.
- The "**Lost in the Middle**" Problem: LLMs tend to focus on the beginning and end of long contexts while neglecting the middle, leading to incomplete reasoning.
- Post-retrieval techniques like **Re-Ranking** and **Context Selection/Compression** help optimize retrieved content before passing it to the LLM.

Source:

- « Lost in the Middle: How Languages Models Use Long Contexts », Liu et al., <https://arxiv.org/abs/2307.03172>
- RAG for LLMs: a Survey, Gao et al., <https://arxiv.org/pdf/2312.10997>

Post-Retrieval

Re-Ranking

Purpose

Reorders document chunks to **highlight the most pertinent results first**, acting as both a **refinement and filtering mechanism**.

Key Techniques

1. **Rule-based** Re-Ranking: Uses predefined metrics such as Diversity, Relevance, and MRR.
2. **Model-based**:
 - Encoder-Decoder models
 - Learning-to-rank models
 - Prompt Engineering + LLMs



Metadata can be utilized in rule-based re-ranking by prioritizing documents based on criteria such as recency or preferred sources.

Post-Retrieval

Context Compression

Purpose

Long contexts introduce noise and dilute key information.

Key Techniques

1. **Token-Level Compression:** Uses Small Language Models to remove unimportant tokens while keeping essential meaning (from a LLM perspective).
2. **Information Extraction & Summarization:** Use models to extract the most useful context and eventually summarize existing contexts.
3. **Prompt Engineering + LLMs**

Source:

- RAG for LLMs: a Survey, Gao et al., <https://arxiv.org/pdf/2312.10997>



Advanced Prompting



Advanced Prompting

Guidelines

Purpose

Creating well-structured prompts is crucial for enhancing the accuracy and effectiveness of LLMs in RAG systems. A well-crafted prompt **acts as a clear guide**, ensuring the LLM understands the task, context, and expected response format.

Key Components of an Effective Prompt

1. **Role:** Specify the function or persona the LLM should assume (e.g., "*You are an AI research assistant.*").
2. **Objective:** Clearly define the purpose of the response (e.g., "*Provide a summary of the retrieved documents.*").
3. **Context:** Include relevant background information to help the model generate informed answers.
4. **Instructions:** Outline the preferred response structure and style (e.g., "*Use bullet points to list key insights.*").
5. **Examples:** Provide sample responses or reasoning patterns to guide the model's output.



Now, platforms for sharing prompts, such as LangChain's Smith Hub, exist. Feel free to draw inspiration from them. <https://smith.langchain.com/hub/>

Advanced Prompting

Guidelines



It's easy to assume that strong prompting skills can enable any model to handle any task, especially when examples significantly enhance the LLM's performance. However, providing examples can sometimes reduce the model's ability to generalize, causing it to struggle with tasks it previously handled well. This is why having a robust evaluation pipeline is essential!



CLOSING WORDS



Getting Started

